Binturong Game
10x dev(s)

Team

Autobiography

My name is Hayden Dennis and I am class of 2027 graduating Fall semester 2026. I am pursuing a career in software engineering with focuses on performant computer vision and efficient data processing. I have worked at Etegent Technologies as a part of UC's COOP program. In addition to the aforementioned skills I have become proficient in fullstack applications and DevOps technologies. In my free time I train Brazilian Jiu Jitsu.

Advisor: Austin Coontz

Austin is an University of Akron graduate and an Offensive Security Professional at TrustedSec here in Ohio. He serves primarily as a penetration testing consultant.

I (Hayden) have known Austin for years from my hometown and have worked together on projects and open source contributions.

Soundtrack

Music for Binturong Game was mastered and produced by Joseph Bowling and Robert Smart.

Stack

- Backend implements FastAPI endpoints for rapid development and is typed via the SQL-ORM SQLModel by Pydantic.
- Postgres is leveraged via docker for the production environment for both object storage and database calls. sqlite3 and s3 containerized via docker were used for local testing and development.
- A Helm managed Kubernetes cluster allows both scaling of the backend service and dynamic creation of game servers.
- Node serves our client which leverages express, Three and sockets in order to maintain a real-time game loop.

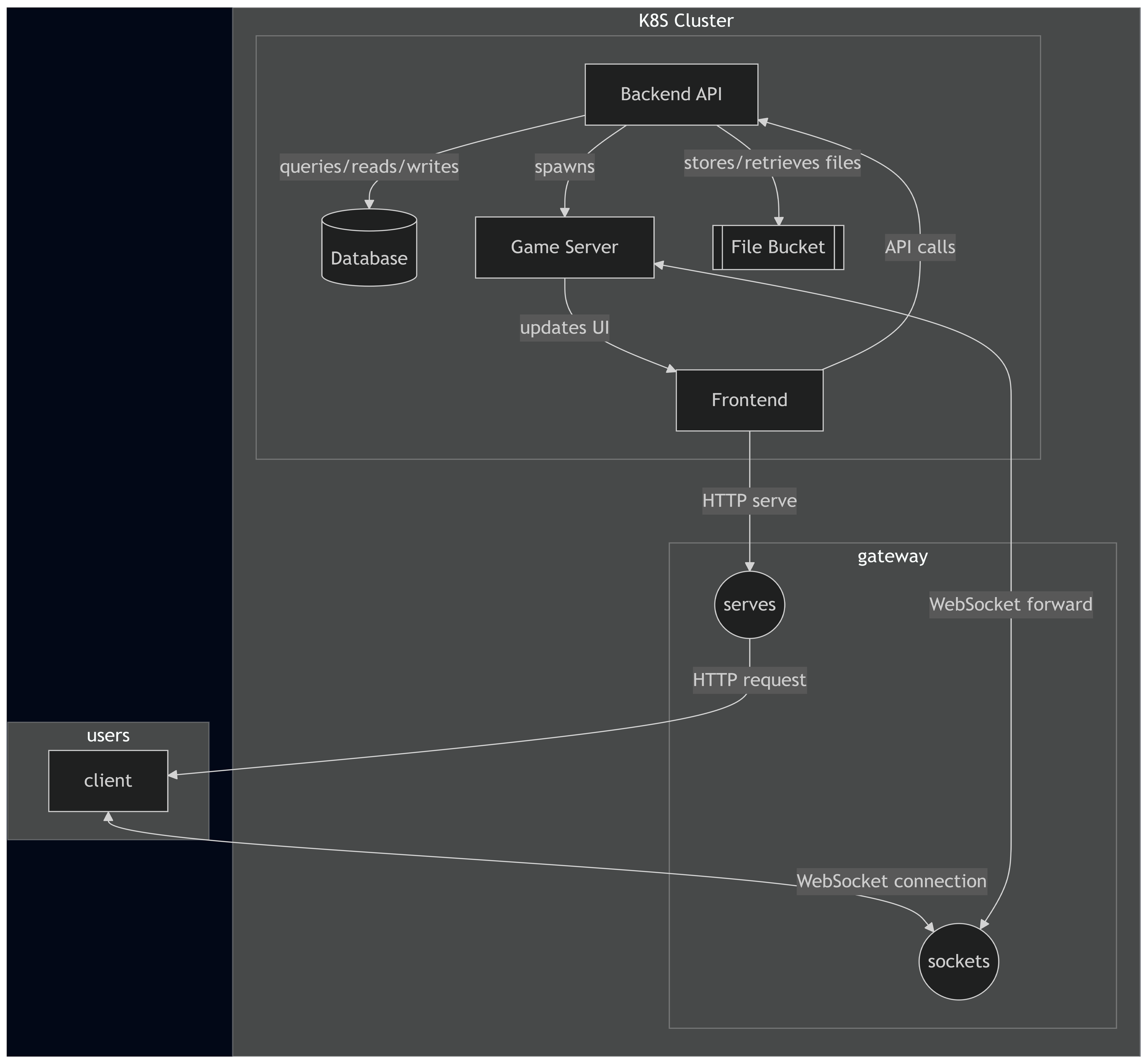
Problem

Existing tooling is not self hostable and locks key features behind paywalls. Existing tools are often proprietary and require the user to work within their ecosystem. There are no existing Anki projects that follow a similar self-hostable tooling design. Other games that take advantage of Anki are less interactive and less fun. Having a public facing URL where a host can create a lobby that is code-locked is crucial for use with educators.

Approach

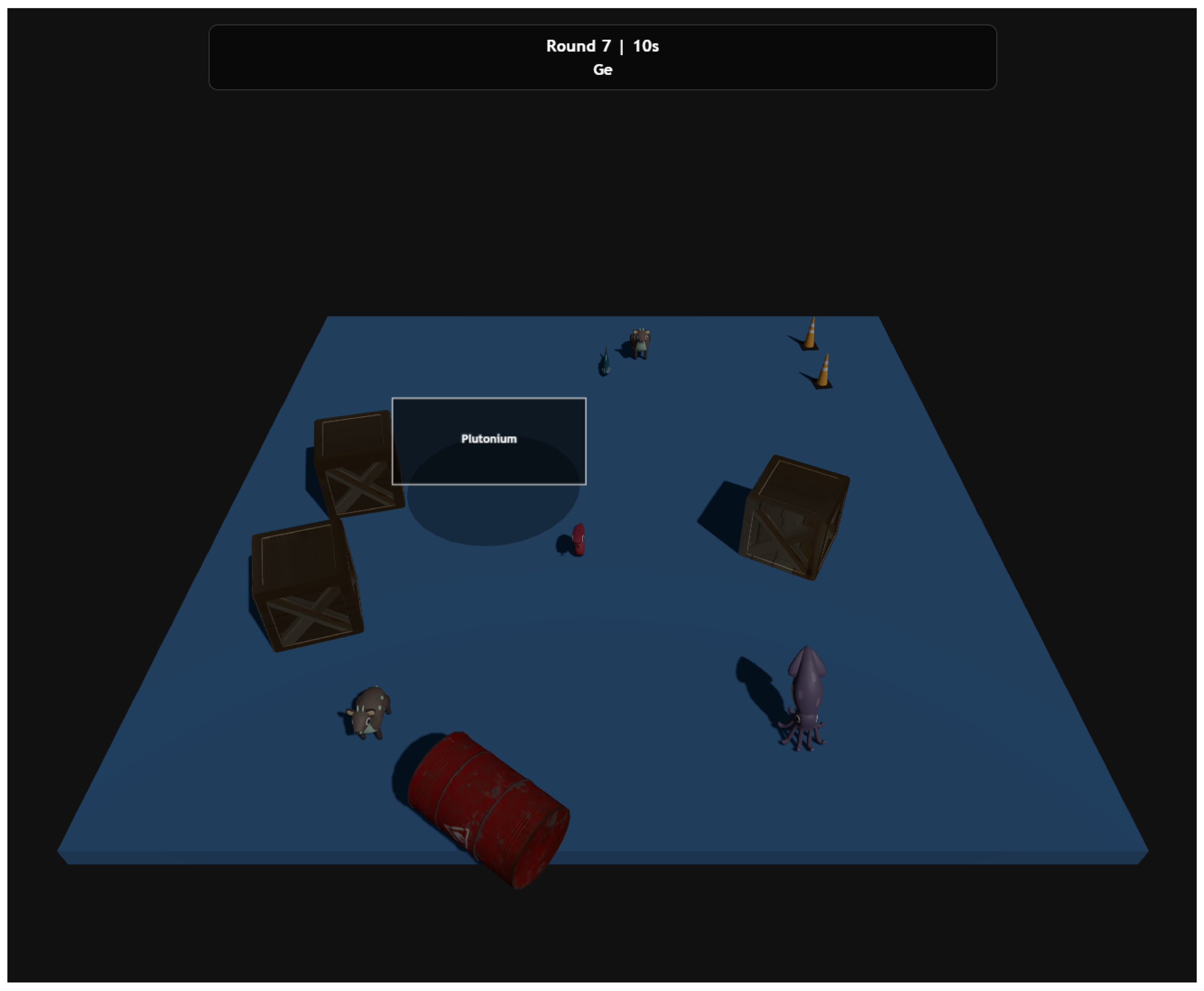
Binturong is designed to be a FOSS game for students to study questions using competition. A key goal is make the solution freeware so anyone can host. The intent is for the game application to provide options for generating card packs via LLM. The distinguishing factor of this project is to encourage competition in order to trick users into studying. The application will be designed to be short review games where players have a score or survival objective. The player's goal will be to control his or her character and select the correct answer based on the game.

Architecture



Future Work

The project had originally planned more features that for time reasons were cut or are still under development. The LLM generation pipeline is the only core feature that remains unfinished. While the team did get results from the file generation pipeline on occasion, the output produced by the LLM often returns broken Anki notes. The main two issues are unembeddable output and nonsense English. Further sanitization and error detection work needs to be developed to create a robust pipeline. The plan for the original game was to include multiple survival scenarios such as parkour and score-based games that were either not included or remain undeveloped. Lastly, a core Anki feature that should be included is support for non-basic notes models. Currently, the game only supports basic flashcard style notes (front/back).



Poster by Hayden Dennis for the UC CEAS Expo